



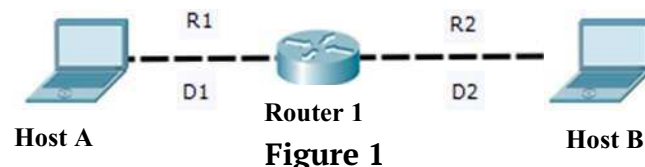
[Any examinee found adopting unfair means will be expelled from the trimester/program as per UIU disciplinary rules.]

- There are **2 (Two)** questions. Answer **both of them**. All questions are of values indicated on the right-hand margin.

Q1.

- a) Suppose, **N users** are connected to a switch that uses **0.5 Gbps** (gigabits per second) link. Each user requires **50 Mbps** and remains active for **20%** of the time.
- What is the **maximum number of users** supported by **circuit switching**? [1]
 - If **packet switching** is used with **15 users**, calculate the **probability** that **11 or more users** are **active** at the same time. [2]
 - Based on the above scenario, **compare** circuit switching and packet switching in terms of their **advantages** and **disadvantages**. [2]

Consider the following diagram (Figure 1) for questions (b)-(d):



- b) Suppose **Host A** and **Host B** are communicating over a **TCP/IP network**. Different **devices (such as hosts, routers)** implement different layers of the TCP/IP model depending on their functionalities.
- Explain** how a layered architecture **simplifies** network operations. [2]
 - Illustrate** the process of **encapsulation** and **de-encapsulation** that happen when a **message M** travels from **A to B** through **Router 1**. [3]
- c) An application on Host A sends a message to Host B using the TCP/IP model. Assume the following:
- Application layer generates **2 Mbytes** of data.
 - Transport layer uses **TCP** with a **20-byte header**.
 - Network layer uses **IP** with a **20-byte header**.
 - Data Link layer adds a **14-byte header** and a **4-byte trailer**.
 - The data link layer supports a **frame size of 1500 bytes (excluding header and trailer)**.
- How many **TCP segments** are required to transmit the data? **What percentage** of the total transmitted bits (at the physical layer) **per segment** is overhead (headers)? [3]
- d) Referring to **Figure 1**, a frame of size **1500 byte** is transmitted from Host A to Host B. Suppose, link capacity **R1 = 100 Mbps**, and **R2 = 1 Gbps**. Link distances are **D1 = 10 meter** and **D2 = 1 kilometer**. Assume the signal **propagation speed** is $1.5 \times 10^8 \text{ m/s}$. The router (Router 1) takes **1 millisecond** to process each packet and there is **no queueing delay** in the router.
- Calculate the **total end-to-end delay** from Host A to Host B. [2]
 - Identify the **bottleneck link** in **Figure 1** and determine the **end-to-end throughput** for a single user. [1]

Q2.

- a) What is a **socket**? Why **port numbers** are used? **Illustrate** with an **example scenario**. [2]
- b) Suppose a webpage has a **base HTML file**, **2 CSS files**, **8 images**. The **bandwidth** between the Client and the Server is **1 Gbps**. The size of each type of file is given below.

Type	Size
HTML	100 KB
CSS	150 KB
Image	2 MB

- The **RTT** for connection setup is **0.2ms**. Find the **total time required** for this webpage to be loaded [3]
for i) **non-persistent HTTP** and ii) **persistent HTTP**.
- c) **Saadman** loves to visit a **static site** (which rarely changes) that contains a lot of information about the galaxy. The server is located at a very distant location and the loading of the webpages is **very slow**. Describe the **possible problem** and show how Saadman can make the loading time of the website faster using **web caching**. **Justify** your answer. [3]
- d) Suppose **Sadiq**, with a **Web-based** e-mail account (**sadiq@hotmail.com**), sends an email to **Reshma** (**reshma@yahoo.com**), who accesses her email using **IMAP**. With a **diagram**, **explain** how the email moves from Sadiq's host to Reshma's host. Be sure to **identify the application-layer protocols** that are used to move the email between the two hosts. [3]
- e) Suppose that a **student (host: lab14.uiu.ac.bd)** from the **computer lab of UIU** is browsing the site <https://scholar.google.com/> and he is the **first person** to browse this site from the UIU network.
- i. **Illustrate with a diagram** how the required IP address of the site will be resolved in case of an **Iterative DNS query**. [2]
- ii. What **drawbacks** you could face if you had used a **recursive query**? [1]

Best Of Luck